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## SEC comment letters and firm disclosure

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## ABSTRACT

In an effort to enhance informational transparency for investors, the SEC periodically reviews public firms' filings for regulatory compliance. Although the SEC dedicates significant resources to the filing review process, the efficacy of this process is unclear. Upon receipt of a comment letter consequent to the SEC's review, the firm can either remedy perceived disclosure deficiencies or attempt to avoid making substantive disclosure changes, including by requesting that certain additional information be treated as confidential or by negotiating with the SEC. In this paper, we first examine the nature, extent, and impact of modifications to firms' disclosures requested by an SEC comment letter. While our initial evidence suggests that firms enhance their disclosure, we find that Rule 406 confidential treatment requests and registrant negotiation have an attenuating effect. Further, consistent with proprietary cost concerns, we show that firms in high tech industries and with greater amounts of R & D are more likely to request confidential treatment. We then examine informational transparency in order to test the SEC's stated intention of the filing review process. We find that improvements to firms' disclosures following a comment letter are associated with a decrease in information asymmetry and a reduction in litigation risk. Collectively, our paper contributes to the literature on disclosure regulation by providing evidence that the SEC comment letter process generally enhances firms' disclosures, improves informational transparency for investors, and mitigates firms' litigation risk, but that some firms take actions that diminish these enhancements.

## 1. Introduction

Securities regulations in the United States are premised on a simple concept: investors should have access to fundamental information about a firm in which they are making an investment. The Securities and Exchange Commission ("SEC") states that "all investors, whether large institutions or private individuals, should have access to certain basic facts about an investment prior to

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<sup>2</sup> Dietrich served as an SEC Academic Fellow in the Office of the Chief Accountant and Johnson was an employee of the SEC's Division of Corporation Finance. Their views expressed here are their own and do not necessarily represent those of the Commission or the staff of the Commission.

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buying it, and so long as they hold it.”<sup>3</sup> To achieve this objective, the SEC issues regulations that require public firms to disclose certain information to the public.<sup>4</sup> Managers must interpret these regulations and prepare corporate reports that disclose information as required by SEC regulations. The SEC issues guidance to assist corporate financial reporting managers in implementing its disclosure regulations, including staff interpretations, speeches or other public statements by Commissioners and staff members.<sup>5,6</sup>

Beyond the provision for general guidance, staff members in the Division of Corporation Finance (“Corp Fin”) systematically review corporate filings such as registration statements and annual or other periodic reports. This filing review process can result in a comment letter requesting that a firm provide additional information so that the staff can better understand the firm’s disclosure.<sup>7</sup> The firm’s response, which can involve multiple rounds of correspondence, could lead the staff to conclude that the filing conforms to SEC regulations in all material respects. However, the staff may determine that a public firm’s filing does not comply with the SEC’s disclosure requirements, for at least two reasons. First, a manager may choose to not disclose certain information and therefore decide to omit information that is required to be disclosed or disclose information in a way that is difficult for (at least some) readers to understand. This view is consistent with Warren Buffett’s statement that “if you can’t understand a footnote or other managerial explanation, it’s usually because the CEO doesn’t want you to.”<sup>8</sup> A second reason is that a manager could interpret and implement a disclosure requirement in a way that differs from what the SEC intended. The staff will offer corrective advice, either to revise a previously filed document or to alter the firm’s disclosure approach in future filings.

Comment letter reviews have been conducted as long as the SEC has existed.<sup>9</sup> Based on its allocation of staff resources, Corp Fin views filing reviews as a substantial activity.<sup>10</sup> Notwithstanding this view, whether or not the comment letter process results in substantive disclosure changes with favorable implications for firms’ informational transparency remains unclear. Despite the SEC’s Compliance and Disclosure Interpretations, speeches and other interpretive guidance, anecdotal evidence suggests that firms are reluctant to make disclosure changes unless prompted by an SEC comment letter. For example, Corp Fin Deputy Director Shelly Parratt notes that “...it seems that many [firms] are reluctant to address these comment themes until we provide specific comments requesting enhanced disclosure...”<sup>11</sup> Although Parratt’s comments imply that firms change their disclosures in response to specific comments from the SEC, Robinson et al. (2011) shows that “non-compliance” may continue due to the costs of verification and limitations in expected penalties. The study further notes that the market does not impose additional penalties because of its inability to discern between an agency motive and a proprietary cost motive of non-compliance with the SEC staff’s directives. As such, the primary question which motivates our study is to what extent the SEC comment letter process prompts firms to modify their disclosures consistent with the SEC’s intent of improving informational transparency for firm stakeholders and whether the disclosure changes are meaningful from the perspectives of investors and information intermediaries.<sup>12</sup>

After receiving a comment letter, a firm can comply with the staff’s directions; alternatively, a firm might attempt to impede the process by making few or even no improvements to their disclosures. A firm that endeavors to minimally comply with the staff’s directions may anticipate that Corp Fin staff will not review the future filing, perhaps due to resource constraints or because it believes that its disclosures sufficiently conform to applicable disclosure requirements. In some instances, if the firm and Corp Fin staff disagree on whether the firm’s filing complies with SEC regulations, Corp Fin staff can refer the firm to the SEC’s Division of Enforcement for further action. However, even if Corp Fin refers a firm, the Division of Enforcement is not obligated to take any further action.<sup>13</sup> In choosing how best to strategically respond to the SEC, firms must weigh the potential benefits from enhanced disclosures against its costs. For example, while the SEC’s requested changes to a firm’s disclosures might result in increased informational transparency for the firm’s stakeholders and/or the potential for decreased expected litigation costs for the firm, it could also reveal sensitive information about the firm that is valuable to the firm’s competitors.

There are at least four ways in which firms navigate the comment letter process without fully complying with all disclosure requests (see Appendix A for examples). First, the firm can simply not respond to an SEC comment letter. While this non-response could result in a referral to Enforcement for further action, the SEC’s budget is limited and thus it may become burdensome for the SEC to sustain legal battles, especially given the number of firms they review. Second, the firm can agree to provide additional disclosure in the future, such as in the next annual report, but then fail to fully incorporate the agreed-upon revisions in the future filing. As the SEC does not have formal procedures in place to consistently follow up before the next periodic review, the deficient

<sup>3</sup> See “How the SEC Protect Investors, Maintains Market Integrity, and Facilitates Capital Formation” at <http://www.sec.gov/about/whatwedo.shtml>.

<sup>4</sup> Public firms include firms with securities traded on a national exchange, or that have more than 500 investors and \$10 million in assets. Public firms often are referred to as “registrants.”

<sup>5</sup> Interpretations include: staff accounting bulletins, compliance and disclosure interpretations, the *Financial Reporting Manual* issued by the Division of Corporation Finance, and other staff reports. See <http://www.sec.gov/interp.shtml>.

<sup>6</sup> See <http://www.sec.gov/news/speech.shtml>.

<sup>7</sup> Fig. 1 depicts trends in annual report reviews and comment letter issuance.

<sup>8</sup> See “Chairman’s Letter,” by Warren E. Buffett, in Berkshire Hathaway’s 2002 Annual Report, page 21.

<sup>9</sup> Evidence of this long-standing review and comment letter process can be found on the SEC Historical Society website ([www.sechistorical.org](http://www.sechistorical.org)) which has references to a “letter of comment” process dating back to the 1930s. Recent SEC Annual Reports document details regarding the number of reviews conducted each fiscal year.

<sup>10</sup> In 2012, approximately 80 percent of Corp Fin’s employees were engaged in the filing review process. See <http://www.sec.gov/divisions/corpfin/cffilingreview.htm>.

<sup>11</sup> See <http://www.sec.gov/news/speech/2009/spch110909sp.htm>.

<sup>12</sup> In a related study, Brown et al. (2017) examines potential comment letter spill-over effects on peer firms that did not receive a comment letter and finds evidence that such “no letter” firms modify their risk factors sections to a greater extent.

<sup>13</sup> Boone et al. (2013) finds no evidence of an increase in the number of Enforcement investigations following an SEC annual report review.

disclosure can persist for multiple years.<sup>14</sup> Third, the firm can negotiate with the staff on the appropriateness of the current disclosures, as many comment letter issues are perceived by firms and staff to be inherently subjective. Comment letters generally begin with solicitation for additional information to better understand the accounting judgments and disclosure choices. Even in cases where the staff originally requests an amendment to the current filing being reviewed, through providing supplemental information on the facts and circumstances considered in arriving at the current disclosure practices, the firm may attempt to satisfy the staff's concerns without modifying current or future disclosures.<sup>15</sup> Fourth, as part of the comment letter process, a firm can try to keep private at least a portion of its response letter by making a confidential treatment request. Unless confidential treatment is requested, the information the firm provides the SEC in its review is not private—the firm knows that any information provided in the comment letter process will eventually become public. Thus, in order to prevent disclosure of information the firm believes is sensitive (e.g., to its competitive position), it could make a confidential treatment request.<sup>16</sup> Some confidential treatment requests are made in the context of supplemental information requested by the SEC staff to assist them in determining the adequacy of the firm's disclosures (e.g., Rule 83 requests). Other confidential treatment requests are associated with disclosures required to be included in the mandatory reports being reviewed by the staff, such as disclosures relating to agreements with suppliers or employees (e.g., Rule 406 requests). This latter category of confidential treatment requests not only redacts the information from the comment letter correspondence, but also may justify omission of the otherwise required information in a current or future filing.

Several academic research surveys have called for more empirical research on the effectiveness of disclosure regulation. Healy and Palepu (2001) raises the concern that “fundamental questions about the...effectiveness of...disclosure regulation in the economy remain unanswered.” Leuz and Wysocki (2016), citing this literature review, states “As Healy and Palepu (2001) note in their survey, empirical research on disclosure regulation...is rare and most of these studies focus on early U.S. disclosure regulation in the 1930s.” The limited amount of empirical research in the area of disclosure regulation is surprising given the central role that regulation plays in shaping the disclosure landscape and thus academic research on disclosure. Leuz and Wysocki (2016) concludes with suggestions for future research, including a better understanding of the “dynamics” and “process” through which financial disclosures are regulated. The survey also suggests an examination of *ex ante* (preventative) approaches to disclosure regulation in addition to *ex post* (enforcement or litigation) approaches. Additionally, Beyer et al. (2010) notes that “not much is known about the effect of regulation on the incentives to produce and disseminate information and the resulting information asymmetry between insiders and outsiders and among investors...”

We seek to partially address this gap by investigating the impact of the SEC comment letter process on firms' disclosures as well as their information environments. We measure modifications to firm's disclosures by extracting five attributes of annual reports that are commonly studied in the qualitative disclosure literature. For the sake of parsimony, we then use a variable reduction technique to collapse the attributes into a single factor, which serves as the basis for our qualitative disclosure proxy. We find that comment letter reviews are associated with the qualitative disclosure proxy as well as four out of the five individual qualitative disclosure attributes used to construct the proxy. We perform validity tests on the univariate significance and direction of the disclosure changes for each of the five disclosure attributes and generally find results consistent with our predictions.

Because the comment letter process is intended to be a dialogue, as noted above, there are a variety of ways in which a firm can respond to a comment letter. We therefore examine how firms' responses to comment letters affect their disclosures. We first document that firms generally modify their disclosures along the dimensions we study following a comment letter review. However, in responding to the comment letter, a firm sometimes supplementally provides information that it views as potentially negatively impacting the firm if the information is disclosed publicly. The firm may determine that providing the sensitive information to the SEC staff in the response letter will facilitate the SEC staff's understanding of issues raised in the comment letter; alternatively, the firm may provide the information because the SEC staff specifically requested the information.

If the firm supplementally provides information in the comment letter process that it views as sensitive, it can request that the SEC staff not disclose the information nor require that the information be disclosed in public filings. Confidential treatment requests therefore provide a mechanism that can limit disclosure changes that might otherwise obtain as a consequence of the comment letter review. We find that, while registrants generally modify their disclosures after receiving a comment letter, those modifications are attenuated for firms that make a Rule 406 confidential treatment request. Moreover, consistent with proprietary cost concerns, we further document that firms in high tech industries and with greater amounts of R&D are more likely to request confidential treatment.

In the absence of a confidential treatment request, the modifications we observe to firms' disclosures are likely a result of those firms that choose not to “push back” on SEC comment letters requesting improvements to disclosures. We therefore explore the effect

<sup>14</sup> To provide some empirical support regarding the regularity of firm non-response and/or non-compliance, we examine the Audit Analytics (AA) database and find that firms *do not respond* to the initial comment letter generated by the SEC in 33% of all comment letter conversations, and firms do not respond to a follow-up letter in 10% of the conversations. We identify non-responders as AA-coded comment letter conversations where there is no response letter to an initial or follow-up SEC letter that is not the “completion of review” letter. In addition, using Altman's Z as a discriminant, we find that only 7% of the non-responders appear to be in financial distress in the year preceding the comment letter. In contrast, the percentage of firms that *do not comply* with the changes requested is quite small at less than 1% of comment letter conversations (AA taxonomy code for “Non-compliance with prior SEC Staff directive”). Regarding the latter statistic, AA's taxonomy code reflects a binary outcome that does not capture the continuum of possible firm responses. All our results are robust when we exclude non-responders.

<sup>15</sup> Certain issues raised by the SEC leave little opportunity for negotiation. For example, if the SEC staff determines that a firm is not in compliance with GAAP on a material accounting judgment, any review outcome other than an amendment of the current filing is unlikely. Nevertheless, even in cases where the initial SEC reviewers take a position that there are deficiencies in the disclosures, managers can appeal to higher levels of authority within the SEC to justify the current disclosures.

<sup>16</sup> See Division of Corporation Finance Staff Legal Bulletin No. 1 “Confidential Treatment Requests”.

of registrant negotiation on the extent of the disclosure modifications. We find that as the number of letters exchanged with the SEC staff increases, improvements to firms' disclosures are again attenuated, which might suggest that firms with substantial legal resources are more effective at lessening the SEC's attempts to influence such firms' disclosures.

Our next set of tests examines the impact of the disclosure improvements on firms' information and litigation environments to provide evidence on the informativeness of the disclosure improvements. We find that qualitative improvements to firms' disclosures prompted by the comment letter process are associated with a decrease in information asymmetry as evidenced by decreased bid-ask spreads. In terms of firm following, we find some evidence of a positive association between analyst coverage and the improvements to firms' disclosures following a comment letter review. Finally, for firms that make large disclosure improvements, we find that their litigation risk is reduced following a comment letter review. Rather than creating information or legal uncertainty, these results suggest that firms that alter their financial disclosures along the dimensions we study benefit from the comment letter process.

In sum, consistent with the calls by Healy and Palepu (2001), Leuz and Wysocki (2016), and Beyer et al. (2010), we add to the literature on disclosure regulation by providing evidence on the role the SEC comment letter process has on enhancing corporate disclosure. Collectively, the evidence presented is consistent with the SEC's objective of influencing firm disclosures as its "chief regulatory tool" in order to enhance informational transparency.<sup>17</sup>

Section 2 details the institutional background of the filing review process. Section 3 provides a literature review and Section 4 lays out specific hypotheses. Section 5 describes the data used in the study and the research design used to test our hypotheses. Sections 6 and 7 provide descriptive statistics and the empirical results, respectively. Section 8 concludes.

## 2. Institutional background

Prior to 2004, individuals, investors, and other interested parties were able to access comment letter correspondence only by request under the Freedom of Information Act ("FOIA"). To provide greater transparency with respect to the comment letter process, in 2004 the SEC decided to publicly release comment letters and response letters after the conclusion of a comment letter review. This decision, and the consequent set of comment (and response) letters, provides public access to details on the timing, content, and outcome of the reviews. The public availability of comment letter correspondence affords an opportunity to investigate the effects of comment letter reviews, which is timely given that the Dodd-Frank Act requires that the SEC annually submit to Congress a report on the effectiveness of its supervisory controls for each of its major divisions. Consistent with the motivation of the current study, the requirement explicitly states that the report include an assessment of the effectiveness of the procedures relating to the "reviews of corporate financial securities filings."<sup>18</sup>

Two of the SEC's largest divisions are Corporation Finance and Enforcement. Each of these divisions has a distinct role in carrying out the SEC's mission of investor protection. Whereas Enforcement focuses on enforcing U.S. securities laws, Corp Fin focuses primarily on monitoring corporate disclosure through its filing review process. Corp Fin reviews two types of filings: registration statements and periodic reports. Registration statements, which are also referred to as transactional filings, include IPOs (Form S-1) and mergers (Form S-4) where the underlying securities are required to be registered under the Securities Act of 1933. In contrast, periodic reports, such as Forms 10-K, 10-Q, 8-K, and proxy statements, are required to be filed under the Securities Exchange Act of 1934.

A filing review can include more than one filing type, such as when a secondary offering filed on Form S-3 incorporates an outstanding 10-K by reference. Transactional filing reviews historically have taken precedence over periodic filing reviews and still have a "timeliness" priority, as they are tied to live transactions. In recent years, reviews of periodic filings occur more frequently and in larger numbers; nearly two-thirds of the comment letters issued between 2004 and 2012 pertain to periodic filing reviews. The significant increase in the number of periodic filing reviews (of which, roughly 77% are 10-K reviews) can be attributed to Section 408 of the Sarbanes Oxley Act of 2002 ("SOX"), which mandates that all SEC registrants must be reviewed at least once every three years and must include a review of Form 10-K. Following the passage of SOX, the SEC's staff size increased. In particular, the number of accounting examiners who have the requisite expertise for 10-K reviews increased markedly. Although periodic filings other than 10-Ks (such as 8-Ks and proxy statements) are reviewed regularly, 10-K reviews are particularly important because 10-Ks provide comprehensive financial disclosures as evidenced by the average burden required to prepare them.<sup>19</sup> Fig. 1 presents the number (Panel A) and percentage (Panel B) of registrants reviewed each SEC fiscal year from 2001 to 2013 along with the number and percentage of comment letters pertaining to 10-K reviews issued since the comment letters began to be publicly released in 2005. Over this time period, roughly 54% of annual reviews result in an initial 10-K comment letter being issued.

The comment letter process is viewed by the SEC as a dialogue with the firm about its disclosure. Upon receipt of a comment letter, the firm is requested to respond within 10 business days or communicate an alternative timeframe. Comments can: (i) request supplemental information to better evaluate the disclosures, (ii) request a revision to future filings, or (iii) request that the filing under review be amended. The SEC has no known process in place to follow up on future filing revisions after the completion of the comment letter process until the firm's next periodic review. For a given comment letter review, the process can vary in the number of comment letter rounds. The comment letters explicitly encourage firms to provide a detailed explanation in their response letter when

<sup>17</sup> Paredes (2003).

<sup>18</sup> See <http://www.gao.gov/products/GAO-13-314>.

<sup>19</sup> As noted by the SEC, the estimated average time burden to prepare a 10-K is about 2000 hours. In contrast, an 8-K requires an average of 6 hours. See <http://www.sec.gov/about/forms/secformsalpha.htm>.

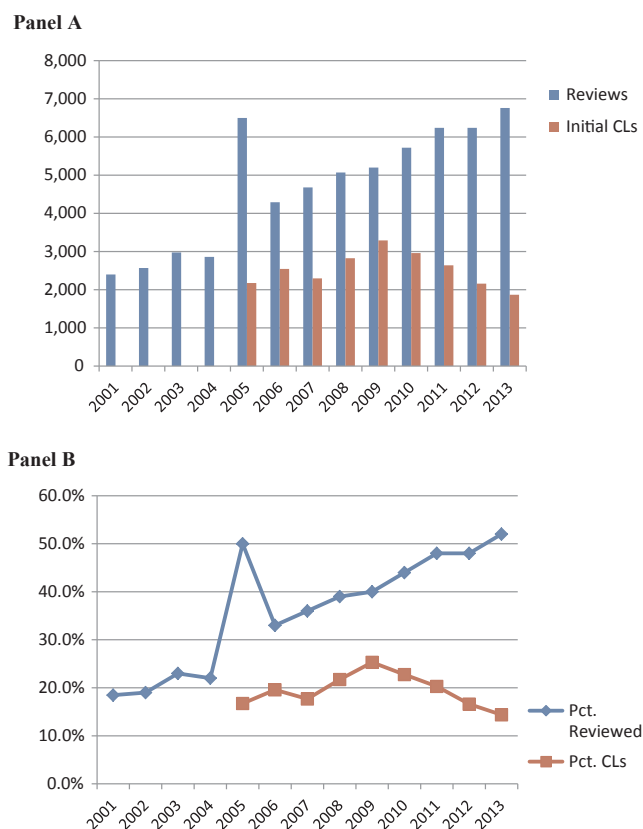


Fig. 1. Panel A (B) of this figure shows the number (percentage) of SEC registrants reviewed each fiscal year as reported in SEC Annual Reports (<http://www.sec.gov/about/secreports.shtml>) and the number (percentage) of annual report comment letters issued each fiscal year per Audit Analytics.

they do not agree with the recommended revisions and to provide supplemental information to help the staff understand their disclosure practices.

At any point during the review, if the firm does not agree with the position or requests of the staff, and chooses to respond to the SEC, the firm can request reconsideration of the issue by the senior level staff of the Assistant Director office, and can also consult with the Office of the Chief Accountant within the Division or ultimately the Office of the Chief Accountant of the SEC. Even in the cases where the staff originally requests an amendment to the filing being reviewed, the firm can choose to negotiate to provide the requested revisions in a future filing or, in some cases, to not revise their disclosures at all.

In more extreme cases, the firm can choose to simply not respond to a comment letter. While non-response may seem risky for the firm, it can be cost-efficient if the firm believes that any perceived deficiencies in their disclosures will not warrant SEC enforcement action. Indeed, the non-response example in Appendix A points to the SEC's use of reputation rather than legal action to enforce compliance (i.e., "As you have not provided a substantive response, we are terminating our review and will take further steps as we deem appropriate. These steps include releasing publicly, through the agency's EDGAR system..."), which is consistent with SEC resource constraints.<sup>20</sup>

In addition, the firm can choose to respond to comments by providing information under a confidential treatment request, where certain portions of the response letter are redacted or supplemental non-public information is subsequently withheld from being filed on EDGAR. There are two categories of confidential treatment requests. Under Rule 83 of the Commission's Rules of Practice, a firm may request confidential treatment with respect to information not required to be filed with the SEC, such as supplemental information only provided in the context of the comment letter review. A common example of a Rule 83 request is for internal management reports provided supplementally to the staff to assess the adequacy of the firm's segment disclosures, which are then returned or destroyed following the completion of the review. The other category of confidential treatment requests falls under Rule 406 of the Securities Act, which relates to information required to be filed with the SEC, such as a material agreement filed as an exhibit to a periodic report.<sup>21</sup> In both cases, the redacted information is not publicly disclosed in the comment letter correspondence.

<sup>20</sup> Consistent with the SEC's implied use of reputation rather than legal action to enforce compliance, we observe in untabulated analysis a negative abnormal market reaction for non-responders on the day of the public comment letter release, and this result holds when we exclude non-responders that have an increased risk of bankruptcy.

<sup>21</sup> See Appendix A for examples of the two types of confidential treatment requests.

nor in current or future filings. The correspondence between the Corp Fin staff and the firm continues until all issues are resolved, at which time, the firm receives a “Completion of Review” letter.<sup>22</sup>

### 3. Prior literature

There is a nascent literature on the determinants and consequences of comment letter reviews. The results of prior studies on the determinants of receiving an SEC comment letter generally are consistent with the requirements in Section 408 of SOX. Cassell et al. (2013) examines firm and auditor characteristics that are associated with a higher likelihood of receiving a comment letter and additionally considers the costs of complying with proposed disclosure modifications. The study finds that firms with larger audit firms and effective internal controls, who are more profitable and less complex, are less likely to receive a comment letter and are able to resolve the comments more quickly. Similarly, Boone et al. (2013) examines the association between accounting standard characteristics and the likelihood and duration of an SEC comment letter review. The study finds that both accounting estimates and rule-based characteristics in GAAP standards increase the probability of receiving a comment letter, and that accounting estimates are positively associated with the duration to resolve such issues. Dechow et al. (2016) examines insider sales prior to the public release of comment letters following the completion of the review and find evidence of strategic trades.

Closely related to our work is Johnston and Petacchi (2017) which finds that bid-ask spreads decline and earnings response coefficients increase around earnings announcements following the completion of 10-K/Q comment letter reviews. Based on changes in market-based measures, the study infers that these changes can be attributed to changes in the firms’ subsequent filings. The study also examines whether or not there is an amended 10-K or 10-Q filed surrounding the comment letter review, which constitutes 17% of the 2004–2006 sample. In contrast, we measure changes in qualitative disclosures along five commonly-studied dimensions following the reviewed 10-K filings, rather than the subsequent earnings announcements. We also explore cross-sectional variation in registrant responses, and examine characteristics of the information environment tied to the disclosures reviewed by the SEC. As such, we complement the findings in Johnston and Petacchi (2017) by directly examining the nature, extent, and impact of qualitative disclosure changes arising from the SEC’s comment letter process and thus contribute to the academic literatures on comments letters and on disclosure regulation more broadly.

### 4. Hypothesis development

#### 4.1. Review impact on annual report disclosures: confidential treatment requests

In preparing each year’s annual report, it is likely that a firm generally maintains the structure and approach it used in its most recently filed annual report. Indeed, a firm can follow the same disclosure practices by preparing its current annual report based on its previous annual report, with revisions only to reflect changes in facts and circumstances between the previous and current years. Such changes might have relatively little impact on certain features of the annual report, such as its length or numerical intensity.

Because firms have incentives to not provide additional or revised disclosure (Verrecchia and Weber, 2006; Robinson et al., 2011), the extent to which the SEC comment letter process influences firm disclosures is unclear *ex ante*. As part of the comment letter process, the firm can submit a confidential treatment request that certain “proprietary” or “sensitive” information in its response be disclosed only to SEC staff. Unless disapproved, the sensitive information is then redacted from the firm’s response letter that is eventually made public as well as the underlying filings being reviewed. Thus, a confidential treatment request (“CTR”) can be viewed as a signal that certain information that the firm or the SEC staff views as facilitating interpretation of the firm’s current disclosure is sensitive and that the firm does not want to disclose this information publicly. This is consistent with Robinson et al. (2011) which discusses how disclosure “non-compliance” can be a possible equilibrium due to the costs of verification and limitations in expected penalties. As we are interested in how the CTR relates to future disclosures, we focus on those CTRs related to Rule 406, where the information is required to be filed with the SEC (i.e., as opposed to Rule 83 CTRs, where the information is provided to the SEC only within the confines of the comment letter). We state our first hypothesis as follows:

**H1.** The impact of comment letter reviews on firms’ disclosures is attenuated for firms making a Rule 406 confidential treatment request.

#### 4.2. Review impact on annual report disclosures: negotiation

Comment letters explicitly encourage firms to provide a detailed explanation in their response letters when they do not agree with the recommended revisions and to provide supplemental information to help the staff understand their disclosure practices. Consequently, a firm can negotiate disclosure improvements that it is willing to make with the SEC staff. If the firm and initial review team are unable to come to a resolution on a given disclosure issue, the firm can appeal to higher levels of authority within the SEC. We presume that a higher level of “push back” or appeals is associated with a higher number of comment letter rounds. Therefore, the extent of observed disclosure improvements likely depends on the “success” of the negotiation process from the firm’s perspective; that is, the firm’s ability to challenge the SEC staff through negotiation in order to minimize requested disclosure changes. We state

<sup>22</sup> For a summary of the SEC’s Filing Review Process, see [www.sec.gov/divisions/corpfin/cffilingreview.htm](http://www.sec.gov/divisions/corpfin/cffilingreview.htm).

our second hypothesis as follows:

**H2.** The impact of comment letter reviews on firms' disclosures is attenuated by the extent of negotiation.

#### 4.3. Firms' information environment, information intermediary following, and litigation risk

Prior literature has examined how disclosure changes affect firms' information environments (e.g., Miller, 2010; Lehavy et al., 2011). We extend these studies to examine the comment letter setting in order to assess the real effects of disclosure changes associated with comment letter reviews. A comment letter review has implications for a firm's information environment if the resultant disclosure improvements reduce existing information asymmetries among investors (Amiram et al., 2016; Leuz and Wysocki, 2016). Changes in a firm's disclosure practices could decrease information asymmetry if improvements to the firm's disclosures reduce investors' differential private information and/or processing abilities. For example, a change in a firm's disclosures requested by the SEC could come in the form of requests for missing disclosure (such as pro forma financials for discontinued operations, income tax disclosures, mining property descriptions, material contracts), revisions to misleading disclosure (for example, a required revision to an estimate of the amount of loss from uncollectible receivables), or enhanced detail or clarity to existing disclosure (e.g., to goodwill impairment estimates, segment reporting, or the presentation of non-GAAP measures). Despite the foregoing discussion, given the rich disclosure environments that exist among SEC reporting firms, the quality of accounting standards in the U.S., and oversight by auditors and other regulators, an SEC review may not result in a material change to a firm's information environment. We state our third hypothesis as follows:

**H3.** Improvements to firms' disclosures arising from comment letter reviews alter firms' information environments.

Next, the decision by information intermediaries to cover a firm might be impacted by improvements to the firm's disclosures following a comment letter review (Bhushan, 1989; Lang and Lundholm, 1996). Revelation of financial information or perceived changes in informational transparency possibly will attract greater attention to a firm, especially if equity analysts who are considering initiating coverage believe that disclosure improvements following a comment letter review indicate a firm's commitment to provide more transparent disclosure (Lehavy et al., 2011). Alternatively, the comment letter process might reveal disclosure deficiencies, which could be interpreted as managers' reluctance to be forthright with the public, thus prompting a decrease in analyst following. We state our fourth hypothesis as follows:

**H4.** Improvements to firms' disclosures arising from comment letter review alter analyst following.

Lastly, if firms make improvements to their disclosures aligned with the SEC's intent of improving informational transparency, such actions likely reflect firms' commitments towards more transparent disclosure following a comment letter review. Given that securities class-action lawsuits often are spurred by allegations of false or misleading information, improvements arising out of the comment letter process will deter possible future litigation (Field et al., 2005; Rogers et al., 2011). We state our fifth and final hypothesis as follows:

**H5.** Improvements to firms' disclosures arising from comment letter reviews reduce litigation risk.

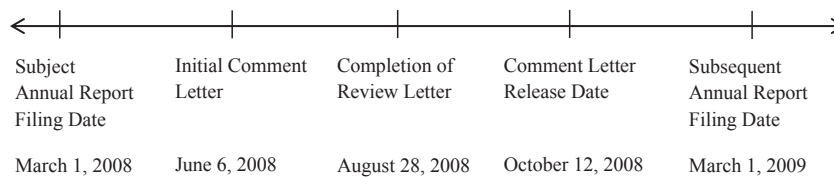
## 5. Data and methodology

### 5.1. Data sources and collection

We use the Audit Analytics ("AA") database to identify and assemble our sample of comment letters. Considering that annual reports are the disclosures most frequently demanded from EDGAR (Drake et al., 2015), annual report reviews comprise 77% of all periodic filing reviews, and annual report reviews are explicitly mandated under SOX 408, we focus exclusively on annual report reviews. Therefore, we retain all initial comment letters in a conversation<sup>23</sup> that focus exclusively on annual report reviews, resulting in 12,955 initial comment letters for the 2004–2011 period. We purge any duplicate comment letter conversations where an initial comment letter is tied to more than one AA-assigned conversation identification code. This screen removes 141 comment letters. Limitations in availability of data used as control variables from CRSP, Compustat, I/B/E/S, and Thomson Financial result in 3105 comment letter conversations in the sample. Next, we identify the annual report that is the subject of the review ("the subject annual report"). We also obtain the subsequent annual report ("the subsequent annual report") to determine qualitative disclosure changes between annual reports. See Fig. 2 for a typical timeline of a comment letter review.

We use Perl to locate and download the subject and subsequent annual reports. Once the subject annual reports are identified from AA, we identify the subsequent annual reports from EDGAR's index files and download both annual reports from EDGAR. This process results in 5804 annual reports corresponding to the 3105 comment letters identified above. Note that we do not obtain precisely twice the number of annual reports as comment letters due to comment letter reviews in consecutive years. Once both the subject and subsequent annual reports have been downloaded, we remove all HTML (or similar) code, embedded files and tables to produce a "flat" text file. Using propensity score matching to address endogeneity concerns, we then match the annual reports subject

<sup>23</sup> A conversation refers to the series of communications between a firm and Corp Fin arising from an initial comment letter.



**Fig. 2.** This figure presents a typical timeline for a comment letter review. The subject and subsequent annual report filing dates spanning the comment letter review are the key event dates of this study.

**Table 1**  
Sample construction.

| Description of selection criteria                                                                                                                      | Observations |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 1. Identify initial Annual Report Comment Letters (“CLs”) in Audit Analytics (2004–2011) and remove duplicate “initial” CLs in the conversation thread | 12,814       |
| 2. Merge CIK to GVKEY and PERMNO                                                                                                                       | 5730         |
| 3. Obtain complete set of available control variables, measured in the period prior to the comment letter filing review                                | 3105         |
| 4. Obtain subject and subsequent annual reports corresponding to CLs in #3 above                                                                       | 5804         |

Table 1 describes the construction of the comment letter sample used in the study.

to comment letter reviews to annual reports for which no comment letter was issued. We match one-to-one using the nearest neighbor on the dimensions of industry, year, and the following SOX 408 criteria in accordance with the [Cassell et al. \(2013\)](#) determinants model: size, return volatility, reported material weaknesses, and restatements. Untabulated analysis suggests that the covariates are well balanced.<sup>24</sup> See [Table 1](#) for the details of the sample construction.

## 5.2. Disclosure proxy

We measure several qualitative disclosure attributes using a firm’s subject and subsequent annual reports to examine improvements to qualitative disclosure associated with comment letter reviews. Following prior qualitative disclosure literature, we examine five attributes in our study: *Length*, *Readability*, *Tone*, *Numerical Intensity* (*NI*), and *Forward-Looking Statement Intensity* (*FLSI*). *Length* and *Readability* are captured, following [Li \(2008\)](#), by the *Lingua::EN::Fathom* module in Perl. *Length* is measured as the number of sentences contained within the annual report. *Readability* is measured using the Fog Index of the annual report ([Gunning, 1952](#)). Note that a higher *Readability* score indicates that the disclosure is *less* readable. *Tone* is measured as the number of optimistic words less pessimistic words contained within the annual report, using the dictionaries developed by [Henry \(2008\)](#) and [Henry and Leone \(2015\)](#) for use in financial discourse. *NI* is measured as the count of non-date numbers in the annual report. *FLSI* is measured as the number of forward-looking sentences contained within the annual report. We identify forward-looking sentences as those containing forward-looking terms and phrases as identified in [Li \(2010\)](#).

We assume that the SEC considers that improved qualitative disclosure is associated with increased *Length*, decreased *Readability* (i.e., lower FOG Index), decreased *Tone*, increased *NI* and increased *FLSI*.<sup>25</sup> Our assumption is predicated on investors’ demand for more information that can be easily processed and the SEC’s response to that demand. Namely, there are several SEC staff speeches and other explicit SEC mentions that reference improvements to qualitative disclosure along the dimensions we study. For example:

- **Length.** A speech by then-SEC Chair Mary Jo White described the history of changes to SEC rules and guidance that have contributed to the expanding quantity of financial disclosures over the years.<sup>26</sup>
- **Readability.** The SEC’s handbook discusses its Plain English initiative.<sup>27</sup>
- **Tone.** Presenting an overly optimistic picture of operating results or liquidity was a concern underlying Reg. G and the interpretive guidance related to non-GAAP measures.<sup>28</sup>
- **Numeric Intensity.** The XBRL mandate resulted in firms disclosing more numeric terms.<sup>29</sup>

<sup>24</sup> There is only one covariate, return volatility, where the difference between treatment and control observations is marginally significant (p-value of 0.10). Our inferences remain unchanged when we re-run all of our tests to include return volatility as an additional control. In untabulated analysis, our results and inferences remain unchanged when we also include M & A, High Tech, and R & D as additional covariates.

<sup>25</sup> See examples in Appendix B.

<sup>26</sup> <https://www.sec.gov/News/Speech/Detail/Speech/1370539878806>.

<sup>27</sup> <https://www.sec.gov/pdf/handbook.pdf>.

<sup>28</sup> <https://www.sec.gov/rules/final/33-8176.htm> and <https://www.sec.gov/divisions/corpfin/faqs/nongapfaq.htm>.

<sup>29</sup> See, for example, [Blankespoor, 2016](#) for a discussion.

- **Forward-Looking Statement Intensity.** Prior to the 1970s, the SEC prohibited projections / forward-looking statements; however, the SEC later reversed its position to encourage more forward-looking disclosures.<sup>30</sup> Consistent with this change, the PSLRA added safe harbor protections for these types of disclosures. MD & A disclosure guidance, for example, encourages disclosure of known trends and uncertainties.<sup>31</sup>

For the sake of parsimony, we calculate a composite measure of qualitative disclosure as the first factor derived from a factor analysis of our five qualitative disclosure attributes discussed above. We call this summary proxy for composite qualitative disclosure the qualitative disclosure factor (*QDF*). In applying the variable reduction technique, we multiply *Readability* and *Tone* by -1 before extracting the factor in order to consistently interpret an increase as an improvement in the disclosure from the SEC's perspective (that is, more readable and less optimistic disclosures are considered as “better” in comment letters reviews). To ensure the attributes which comprise the proxy align with the SEC's intent, we also provide univariate validation tests for each of our qualitative disclosure attributes.

### 5.3. Information environment, information intermediary following, and litigation risk proxies

We utilize a standard information environment proxy, abnormal bid-ask-spread (*BAS*), to examine changes in information asymmetry corresponding to the annual report filing subsequent to the comment letter review relative to the subject annual report. We calculate this measure as the event period average daily percent spread minus the pre-period average daily percent spread. The event period is trading days (0, +2) relative to the annual report filing date and the pre-period is the 45 trading days prior (−45, −1). Daily percent spread is obtained from CRSP and calculated as the daily high offer price less the daily low bid price scaled by the midpoint, all multiplied by 100, consistent with Amiram et al. (2016). Next, we examine changes in analyst following surrounding the subject and subsequent annual report to assess the impact of qualitative disclosure changes on information intermediaries. Analyst coverage (*Analyst\_Cov*) is the average number of analysts following a firm in the 30 days after the annual report filing date. Lastly, we examine litigation risk using data on securities class-action lawsuits from the Stanford Securities Class Action Clearinghouse database. Litigation risk (*Litigation*) is an indicator variable for a firm undergoing a securities class-action lawsuit within a year following the completion of a comment letter review.

### 5.4. Empirical design

We employ the following baseline empirical specification to establish the impact of the comment letter process on firms' disclosures before proceeding to test our hypotheses.

$$QDF = \beta_0 + \beta_1 CL + \beta_2 Post + \beta_3 CL*Post + \beta_4 LogSize + \beta_5 BTM + \beta_6 LogAssets + \beta_7 ROA + \beta_8 Loss + \beta_9 IO + \beta_{10} Analysts + \beta_{11} StdevEarn + \beta_{12} Age + \beta_{13} Beta + \beta_{14} M \& A + \beta_{15} Restate + \beta_{16} CLComplexity + \beta_{17} DisclosureTrend + \varepsilon \quad (1)$$

The unit of analysis is the entire annual report filing corresponding to the comment letter review. In order to isolate the impact of the comment letter review on the annual report, we implement a matched, pre-post design to reflect the subject and subsequent annual reports, respectively. *CL* is an indicator variable for the annual report that is the subject of the review. *Post* is an indicator variable for the annual report subsequent to the report that was the subject of the review. The coefficient of interest,  $\beta_3$ , on the interaction term *CL\*Post* reflects the improvements to the qualitative disclosure factor (*QDF*) for firms subject to a comment letter review relative to matched firms that were not.

All control variables are measured in the accounting period prior to the initial comment letter and are defined in Appendix C. Ten control variables (*LogSize*, *BTM*, *LogAssets*, *ROA*, *Loss*, *IO*, *Analysts*, *Stdev\_Earn*, *Age*, *Beta*) in the model relate to fundamental firm characteristics that are likely associated with qualitative disclosure attributes; four controls (*M & A*, *Restate*, *CL\_Complexity*, *Disclosure\_Trend*) relate to other characteristics that likely also impact qualitative disclosure attributes. For example, being involved in mergers and acquisitions (*M & A*) or a restatement (*Restate*) in the year of or year prior to the subject annual report could impact both the probability of being reviewed by the SEC and the annual report disclosure properties describing these events (Cassell et al., 2013). *CL\_Complexity* is the total number of words in the SEC comment letters.<sup>32</sup> *Disclosure\_Trend* is the average industry-year length of annual reports, which controls for industry disclosure trends, given the concern that annual reports have increased in length over time. In all regression tests, we include industry fixed effects and t-statistics are calculated with standard errors clustered by firm.

For our first and second hypotheses we augment Eq. (1) to include interaction terms for Rule 406 confidential treatment requests and negotiation as follows:

$$QDF = \beta_0 + \beta_1 CL + \beta_2 Post + \beta_3 CTR406 + \beta_4 CL*Post + \beta_5 CL*CTR406 + \beta_6 Post*CTR406 + \beta_7 CL*Post*CTR406 + \beta_x Controls + \varepsilon \quad (2)$$

<sup>30</sup> <https://www.sec.gov/rules/concept/fwdinfo.txt>.

<sup>31</sup> <https://www.sec.gov/rules/interp/33-8350.htm>.

<sup>32</sup> Our inferences remain unchanged when we include an alternate measure of comment letter complexity, the number of filings reviewed by the SEC in the inquiry.

$$QDF = \beta_0 + \beta_1 CL + \beta_2 Post + \beta_3 Negotiation + \beta_4 CL*Post + \beta_5 CL*Negotiation + \beta_6 Post*Negotiation + \beta_7 CL*Post*Negotiation + \beta_x Controls + \varepsilon \quad (3)$$

where *QDF*, *CL*, *Post*, and *Controls* are as defined above. *CTR406* is an indicator variable for a Rule 406 confidential treatment request by the registrant as part of the comment letter process. We obtained from the Audit Analytics database all comment letter responses that include confidential treatment. We then perform a keyword search for “Rule 406” and categorize these CTRs by setting *CTR406* equal to 1 and all other observations to 0. *Negotiation*, which proxies for the propensity of a firm to appeal the SEC staff’s requests during the comment letter process, is an indicator for comment letter reviews with an above-median number of rounds where rounds is the number of SEC comment letters issued by the SEC during a review, excluding the “Completion of Review” letter (also obtained from Audit Analytics database). The coefficient of interest,  $\beta_7$ , on the interaction term *CL\*Post\*CTR406* in Eq. (2) reflects the differential improvements in the qualitative disclosure factor (*QDF*) for reviews where the firm requests confidential treatment under Rule 406. The coefficient of interest,  $\beta_7$ , on the interaction term *CL\*Post\*Negotiation* in Eq. (3) reflects the differential improvement to the qualitative disclosure factor (*QDF*) for reviews where the firm negotiates with the SEC, or in other words, achieves a lower level of disclosure improvements by prolonging the number of comment letter rounds.

To test our third and fourth hypotheses, we modify Eq. (1) to replace the dependent variable with either the information environment or information intermediary proxy and then include an interaction term for the qualitative disclosure factor (*QDF*) as follows:

$$InformationEnvironment \text{ or } InformationIntermediaryFollowing = \beta_0 + \beta_1 CL + \beta_2 Post + \beta_3 QDF + \beta_4 CL*Post + \beta_5 CL*QDF + \beta_6 Post*QDF + \beta_7 CL*Post*QDF + \beta_x Controls + \varepsilon \quad (4)$$

where the information environment construct is proxied for by abnormal bid-ask-spread (*BAS*) and the information intermediary following construct is proxied for by analyst coverage (*Analyst\_Cov*) and are as defined above in Section 5.3 and in Appendix C. *Controls* are the same as those found in Eq. (1) with the exception of the exclusion of prior analyst following (*Analysts*) when analyst coverage (*Analyst\_Cov*) is the dependent variable. The coefficient of interest,  $\beta_7$ , on the interaction terms *CL\*Post\*QDF* in Eq. (4) reflects the change in the firm’s information environment or information intermediary following, respectively, for firms that improve their disclosure following a comment letter review.

To test our fifth and final hypothesis, we modify Eq. (1) to replace the dependent variable with the litigation risk proxy (*Litigation*). We then include an interaction term for firms with the largest (upper quartile) change in the qualitative disclosure factor ( $\Delta QDF$ ) as follows:

$$Litigation = \beta_0 + \beta_1 CL + \beta_2 LogSize + \beta_3 BTM + \beta_4 LogAssets + \beta_5 ROA + \beta_6 Loss + \beta_7 IO + \beta_8 Analysts + \beta_9 StdevEarn + \beta_{10} Age + \beta_{11} Beta + \beta_{12} M \& A + \beta_{13} Restate + \beta_{14} CLComplexity + \beta_{15} DisclosureTrend + \varepsilon \quad (5)$$

$$Litigation = \beta_0 + \beta_1 CL + \beta_2 \Delta QDF + \beta_3 CL*\Delta QDF + \beta_4 LogSize + \beta_5 BTM + \beta_6 LogAssets + \beta_7 ROA + \beta_8 Loss + \beta_9 IO + \beta_{10} Analysts + \beta_{11} StdevEarn + \beta_{12} Age + \beta_{13} Beta + \beta_{14} M \& A + \beta_{15} Restate + \beta_{16} CLComplexity + \beta_{17} DisclosureTrend + \varepsilon \quad (6)$$

## 6. Descriptive statistics

Table 2, Panel A provides descriptive statistics for the comment letter sample used in the empirical analyses. The average annual report contains 2417 sentences. The average readability is greater than 20, indicating that more than 20 years of formal education is required to comprehend the average annual report. The average number of optimistic words net of pessimistic words is approximately 89. The number of numbers is 1248 for the average annual report and the number of forward-looking statements is 787. With respect to the information environment proxies, the average abnormal bid-ask-spread in the three-day event window of the annual report filing date is 54 basis points. As for the coverage proxy, the average number of analysts following firms in the 30 days after the annual report filing date is 5.6. Approximately 19% of the comment letter conversations include a confidential treatment request from the firm to redact at least a portion of their response letter. About 6.5% of the CTRs (1% of the overall sample) are related to Rule 406. The mean (median) number of comment letter rounds is 1.5 (1.0). The firms in our sample are fairly large, with a market capitalization of roughly \$1.2 billion and assets of \$1.4 billion. The average firm is slightly riskier than the market with a beta of 1.13, and 29% of the firms experienced a loss in the most recent fiscal quarter prior to receiving a comment letter. Consistent with the sample being comprised of large firms, both institutional ownership and analyst following are fairly high at 82% and 11, respectively. Approximately 11% (47%) of the firm-years involved a restatement (M & A activity where the firm was either an acquirer or target) during the year of or year prior to the subject annual report. Panel B of Table 2 presents differences in means for the qualitative disclosure factor and its attributes before and after the receipt of a comment letter. Table 3 presents correlation coefficients; the relations are generally as expected.

**Table 2**  
Descriptive statistics.

| Panel A                                                      |       |        |            |           |             |             |
|--------------------------------------------------------------|-------|--------|------------|-----------|-------------|-------------|
| Variables                                                    | N     | Mean   | Median     | Std. Dev. | 25th Pctile | 75th Pctile |
| <i>Qualitative disclosure factor and attributes</i>          |       |        |            |           |             |             |
| QDF                                                          | 5804  | 0.217  | −0.012     | 0.945     | −0.347      | 0.514       |
| Length                                                       | 5804  | 2417   | 2130       | 1321      | 1625        | 2867        |
| Readability                                                  | 5804  | 20.3   | 19.8       | 2.4       | 19.0        | 20.8        |
| Tone                                                         | 5804  | 89.2   | 84.0       | 135.5     | 16.0        | 159.0       |
| NI                                                           | 5804  | 1248   | 1102       | 627       | 859         | 1455        |
| FLSI                                                         | 5804  | 787    | 632        | 581       | 458         | 933         |
| <i>Information environment and litigation risk proxies</i>   |       |        |            |           |             |             |
| BAS                                                          | 4653  | 0.0054 | 0.0004     | 0.0272    | −0.0070     | 0.0114      |
| Analyst_Cov                                                  | 4117  | 5.63   | 4.50       | 4.47      | 2.33        | 7.75        |
| Litigation                                                   | 3197  | 0.033  | 0.000      | 0.177     | 0.000       | 0.000       |
| <i>Conditioning variables</i>                                |       |        |            |           |             |             |
| CTR                                                          | 5744  | 0.186  | 0.000      | 0.389     | 0.000       | 0.000       |
| CTR406                                                       | 5744  | 0.012  | 0.000      | 0.108     | 0.000       | 0.000       |
| Negotiation                                                  | 4881  | 0.360  | 0.000      | 0.480     | 0.000       | 1.000       |
| <i>Control variables</i>                                     |       |        |            |           |             |             |
| CL_Complexity                                                | 5804  | 1690   | 1425       | 1046      | 1179        | 1836        |
| LogSize                                                      | 5804  | 7.054  | 6.970      | 1.849     | 5.708       | 8.340       |
| BTM                                                          | 5804  | 0.603  | 0.488      | 0.871     | 0.272       | 0.777       |
| LogAssets                                                    | 5804  | 7.208  | 7.173      | 1.954     | 5.741       | 8.558       |
| ROA                                                          | 5804  | −0.002 | 0.009      | 0.099     | −0.003      | 0.021       |
| Loss                                                         | 5804  | 0.292  | 0.000      | 0.455     | 0.000       | 1.000       |
| IO                                                           | 5804  | 0.815  | 0.950      | 0.250     | 0.691       | 1.000       |
| Analysts                                                     | 5804  | 10.51  | 9.00       | 7.71      | 5.00        | 15.00       |
| Stdev_Earn                                                   | 5804  | 93.22  | 15.24      | 393.98    | 4.72        | 52.40       |
| Age                                                          | 5804  | 19.99  | 16.75      | 12.84     | 10.75       | 26.75       |
| Beta                                                         | 5804  | 1.132  | 1.088      | 0.530     | 0.777       | 1.434       |
| M & A                                                        | 5804  | 0.473  | 0.000      | 0.499     | 0.000       | 1.000       |
| Restate                                                      | 5804  | 0.112  | 0.000      | 0.316     | 0.000       | 0.000       |
| Disclosure_Trend                                             | 5804  | 10.53  | 10.52      | 0.35      | 10.34       | 10.70       |
| <i>Panel B: Differences in means pre-post comment letter</i> |       |        |            |           |             |             |
| Variables                                                    | Pre   | Post   | Difference | t-stat    |             |             |
| <i>Qualitative disclosure factor and attributes</i>          |       |        |            |           |             |             |
| QDF                                                          | 0.189 | 0.241  | 0.052      | 2.1       | **          |             |
| Length                                                       | 2377  | 2452   | 76         | 2.2       | **          |             |
| Readability                                                  | 20.3  | 20.3   | 0.0        | 0.5       |             |             |
| Tone                                                         | 90.8  | 87.8   | −2.9       | −0.8      |             |             |
| NI                                                           | 1230  | 1265   | 35         | 2.1       | **          |             |
| FLSI                                                         | 776   | 797    | 20         | 1.3       |             |             |

**Table 2**, Panel A provides descriptive statistics for the comment letter sample used in the empirical analyses. Panel B provides differences in means for the qualitative disclosure factor and its attributes before and after the receipt of a comment letter. Please refer to Appendix C for variable definitions.

## 7. Empirical results

### 7.1. Comment letters and annual report disclosure modifications

#### 7.1.1. The qualitative disclosure factor

**Table 4** presents regression results from estimating Eq. (1) on the association between comment letter reviews and differences in the qualitative disclosure factor (*QDF*) between the subject and subsequent annual reports. The coefficient of interest,  $\beta_3$ , on the interaction term *CL\*Post* reflects the improvement to the qualitative disclosure factor (*QDF*) for firms subject to a comment letter review relative to firms that were not subject to review. Here and in subsequent tables, we first present the regression results for the comment letter sample by itself in Column 1. We find a significantly positive change in *QDF*, which provides initial evidence supporting the view that firms modify their disclosure consequent to a comment letter review. The coefficient of 0.059 represents a 27% increase in the mean qualitative disclosure factor. Further, the result is robust in the matched, pre-post model found in Column 2.<sup>33</sup> Overall, these results are consistent with standard language contained in the SEC comment letters to firms: “Please understand that

<sup>33</sup> When we examine the components of *QDF* individually, we find a statistically significant change in the anticipated direction for four of the five qualitative disclosure attributes. On average, these changes result in a 3% increase in disclosure length, a 6% decrease in disclosure optimism, a 3% increase in numerical intensity, and a 3% increase in forward-looking statements.

**Table 3**  
Correlations.

| Variables        | QDF               | CL Complexity     | Length            | Readability       | Tone              | NI                | FLSI              | LogSize           | BTM               | LogAssets         | ROA               | Loss              | IO                | Analysts          | Stdev_Earn        | Age              | Beta              | M & A             | Restate           | Disclosure Trend |
|------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|-------------------|-------------------|-------------------|------------------|
| QDF              | 1                 |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |
| CL_Complexity    | 0.029<br>(0.025)  | 1                 |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |
| Length           | 0.985<br>(0.000)  | 0.025<br>(0.053)  | 1                 |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |
| Readability      | 0.038<br>(0.004)  | 0.025<br>(0.062)  | -0.107<br>(0.000) | 1                 |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |
| Tone             | -0.083<br>(0.000) | -0.015<br>(0.266) | -0.131<br>(0.000) | -0.131<br>(0.000) | 1                 |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |
| NI               | 0.925<br>(0.000)  | 0.041<br>(0.002)  | 0.135<br>(0.000)  | 0.135<br>(0.000)  | -0.083<br>(0.000) | 1                 |                   |                   |                   |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |
| FLSI             | 0.948<br>(0.000)  | 0.023<br>(0.079)  | 0.898<br>(0.000)  | 0.147<br>(0.000)  | -0.142<br>(0.000) | 0.848<br>(0.000)  | 1                 |                   |                   |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |
| LogSize          | 0.297<br>(0.000)  | -0.069<br>(0.000) | 0.309<br>(0.000)  | -0.049<br>(0.000) | 0.112<br>(0.000)  | 0.299<br>(0.000)  | 0.235<br>(0.000)  | 1                 |                   |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |
| BTM              | 0.029<br>(0.030)  | 0.024<br>(0.071)  | 0.025<br>(0.053)  | 0.033<br>(0.012)  | -0.129<br>(0.000) | 0.032<br>(0.014)  | 0.014<br>(0.282)  | -0.226<br>(0.000) | 1                 |                   |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |
| LogAssets        | 0.410<br>(0.000)  | -0.027<br>(0.038) | 0.414<br>(0.000)  | 0.024<br>(0.065)  | -0.042<br>(0.002) | 0.431<br>(0.000)  | 0.316<br>(0.000)  | 0.852<br>(0.000)  | -0.002<br>(0.859) | 1                 |                   |                   |                   |                   |                   |                  |                   |                   |                   |                  |
| ROA              | -0.044<br>(0.001) | -0.051<br>(0.000) | -0.038<br>(0.004) | -0.025<br>(0.054) | 0.075<br>(0.000)  | -0.039<br>(0.003) | -0.046<br>(0.000) | 0.191<br>(0.000)  | -0.095<br>(0.859) | 0.095<br>(0.000)  | 1                 |                   |                   |                   |                   |                  |                   |                   |                   |                  |
| Loss             | 0.035<br>(0.007)  | 0.094<br>(0.000)  | 0.027<br>(0.041)  | 0.031<br>(0.017)  | -0.116<br>(0.000) | 0.036<br>(0.006)  | 0.041<br>(0.002)  | -0.335<br>(0.000) | 0.129<br>(0.000)  | -0.219<br>(0.000) | -0.428<br>(0.000) | 1                 |                   |                   |                   |                  |                   |                   |                   |                  |
| IO               | 0.130<br>(0.000)  | -0.057<br>(0.000) | 0.134<br>(0.000)  | -0.044<br>(0.001) | -0.005<br>(0.691) | 0.124<br>(0.000)  | 0.118<br>(0.000)  | 0.375<br>(0.000)  | -0.031<br>(0.020) | 0.385<br>(0.000)  | 0.016<br>(0.234)  | -0.109<br>(0.000) | 1                 |                   |                   |                  |                   |                   |                   |                  |
| Analysts         | 0.189<br>(0.000)  | -0.051<br>(0.000) | 0.200<br>(0.000)  | -0.056<br>(0.000) | 0.100<br>(0.000)  | 0.163<br>(0.000)  | 0.168<br>(0.000)  | 0.722<br>(0.000)  | -0.134<br>(0.000) | 0.600<br>(0.000)  | 0.082<br>(0.000)  | -0.181<br>(0.000) | 0.330<br>(0.000)  | 1                 |                   |                  |                   |                   |                   |                  |
| Stdev_Earn       | 0.188<br>(0.000)  | 0.020<br>(0.122)  | 0.184<br>(0.000)  | 0.048<br>(0.000)  | -0.031<br>(0.017) | 0.207<br>(0.000)  | 0.146<br>(0.000)  | 0.320<br>(0.000)  | -0.046<br>(0.001) | 0.372<br>(0.000)  | 0.016<br>(0.235)  | -0.020<br>(0.136) | 0.025<br>(0.056)  | 0.256<br>(0.000)  | 1                 |                  |                   |                   |                   |                  |
| Age              | 0.020<br>(0.134)  | -0.058<br>(0.000) | -0.001<br>(0.935) | 0.046<br>(0.001)  | -0.119<br>(0.000) | 0.090<br>(0.154)  | 0.019<br>(0.000)  | 0.366<br>(0.000)  | -0.034<br>(0.010) | 0.396<br>(0.000)  | 0.069<br>(0.000)  | -0.125<br>(0.000) | 0.177<br>(0.000)  | 0.174<br>(0.000)  | 0.164<br>(0.000)  | 1                |                   |                   |                   |                  |
| Beta             | 0.125<br>(0.000)  | 0.021<br>(0.111)  | 0.121<br>(0.000)  | 0.002<br>(0.895)  | -0.079<br>(0.000) | 0.154<br>(0.000)  | 0.098<br>(0.000)  | 0.093<br>(0.000)  | 0.021<br>(0.109)  | 0.178<br>(0.000)  | -0.021<br>(0.102) | 0.100<br>(0.000)  | 0.265<br>(0.000)  | 0.054<br>(0.000)  | 0.049<br>(0.050)  | 0.026<br>(0.050) | 1                 |                   |                   |                  |
| M & A            | 0.082<br>(0.000)  | 0.000<br>(0.972)  | 0.089<br>(0.000)  | -0.012<br>(0.378) | 0.118<br>(0.000)  | 0.079<br>(0.000)  | 0.059<br>(0.000)  | 0.224<br>(0.000)  | -0.016<br>(0.223) | 0.216<br>(0.000)  | 0.011<br>(0.389)  | -0.080<br>(0.000) | 0.110<br>(0.000)  | 0.158<br>(0.000)  | 0.101<br>(0.000)  | 0.078<br>(0.000) | -0.025<br>(0.058) | 1                 |                   |                  |
| Restate          | 0.036<br>(0.006)  | 0.030<br>(0.021)  | 0.039<br>(0.003)  | -0.025<br>(0.061) | -0.012<br>(0.343) | 0.034<br>(0.009)  | 0.029<br>(0.026)  | -0.083<br>(0.000) | 0.000<br>(0.994)  | -0.064<br>(0.000) | -0.033<br>(0.012) | 0.093<br>(0.000)  | -0.047<br>(0.000) | -0.080<br>(0.000) | -0.019<br>(0.142) | 0.027<br>(0.040) | 0.029<br>(0.027)  | -0.031<br>(0.017) | 1                 |                  |
| Disclosure_Trend | 0.226<br>(0.000)  | -0.030<br>(0.024) | 0.225<br>(0.000)  | 0.052<br>(0.000)  | -0.044<br>(0.001) | 0.222<br>(0.000)  | 0.181<br>(0.000)  | 0.154<br>(0.000)  | 0.006<br>(0.650)  | 0.222<br>(0.000)  | -0.035<br>(0.007) | -0.026<br>(0.049) | 0.111<br>(0.000)  | 0.056<br>(0.000)  | 0.042<br>(0.001)  | 0.038<br>(0.004) | 0.026<br>(0.045)  | -0.009<br>(0.502) | -0.023<br>(0.084) | 1                |

Table 3 presents the pairwise correlations between variables used in the study for the comment letter sample. P-values are shown in parentheses below each Pearson correlation coefficient. Please refer to Appendix C for variable definitions.

**Table 4**  
SEC comment letters and changes in the qualitative disclosure factor (QDF).

$$QDF = \beta_0 + \beta_1 CL + \beta_2 Post + \beta_3 CL*Post + \beta_4 LogSize + \beta_5 BTM + \beta_6 LogAssets + \beta_7 ROA + \beta_8 Loss + \beta_9 IO + \beta_{10} Analysts + \beta_{11} Stdev.Earn + \beta_{12} Age + \beta_{13} Beta + \beta_{14} M\&A + \beta_{15} Restate + \beta_{16} CL\_Complexity + \beta_{17} Disclosure\_Trend + \varepsilon$$

| Variables | QDF                  | QDF                   |
|-----------|----------------------|-----------------------|
| Intercept | −3.470***<br>(−5.86) | −9.748***<br>(−16.21) |
| CL        |                      | −0.053**<br>(−2.54)   |
| Post      | 0.059***<br>(3.63)   | −0.028*<br>(−1.86)    |
| CL*Post   |                      | 0.060***<br>(2.71)    |
| N         | 5804                 | 11,770                |
| Controls  | Yes                  | Yes                   |
| R-squared | 0.282                | 0.302                 |

Table 4 presents regression results on the association between the SEC comment letter process and the change in the qualitative disclosure factor. *QDF* is the first factor derived from the five qualitative disclosure attributes. Column 1 presents the results for the comment letter sample. Column 2 presents the results of the matched, pre-post model. t-statistics are shown in parentheses below the regression coefficients and are computed using standard errors clustered by firm. All regressions include industry fixed effects. Please refer to Appendix C for variable definitions.

\* Denotes statistical significance at the 0.10 significance level.

\*\* Denotes statistical significance at the 0.05 significance level.

\*\*\* Denotes statistical significance at the 0.01 significance level.

**Table 5**  
Univariate validation tests for qualitative disclosure attributes.

| Panel A: Full Sample        |      |      |          |
|-----------------------------|------|------|----------|
| ΔDisclosure_Proxy           | N    | Mean | t-stat   |
| ΔLength                     | 7576 | 22.3 | 1.64     |
| ΔReadability                | 7576 | 0.2  | 6.71***  |
| ΔTone                       | 7576 | −8.1 | −6.90*** |
| ΔNI                         | 7576 | 8.7  | 1.29     |
| ΔFLSI                       | 7576 | 12.7 | 1.85*    |
| Panel B: Conditional sample |      |      |          |
| ΔDisclosure_Proxy           | N    | Mean | t-stat   |
| ΔLength                     | 4604 | 31.4 | 1.91*    |
| ΔReadability                | 2468 | 0.2  | 4.92***  |
| ΔTone                       | 2468 | −8.7 | −4.05*** |
| ΔNI                         | 2318 | 21.5 | 1.86*    |
| ΔFLSI                       | 5364 | 20.2 | 2.51**   |

Table 5 presents validation tests for the qualitative disclosure proxy. Panel A presents the univariate significance of the change in each qualitative disclosure attribute between the subject and subsequent annual report for the unrestricted comment letter sample. Panel B presents the univariate results conditioning the sample based on whether the comment letter requests a future filing revision (as a validation test for *Length*) or contains relevant keywords regarding each qualitative disclosure attribute. For *Readability* we condition the sample on comment letters containing the keywords: plain english, boilerplate, clarify, explain, or define. For *Tone* we condition the sample on comment letters containing the keywords: tone, balanced, optimistic, positive, or negative. For numeric intensity (*NI*) we condition the sample on comment letters containing the keywords: quantify or quantitative. Finally, for forward-looking statement intensity (*FLSI*), we condition the sample on comment letters containing the keywords: forward-looking, forward looking, prospective, or future. Please refer to Appendix C for variable definitions.

\* Denotes statistical significance at the 0.10 significance level.

\*\* Denotes statistical significance at the 0.05 significance level.

\*\*\* Denotes statistical significance at the 0.01 significance level.

the purpose of our review process is to assist you in your compliance with the applicable disclosure requirements and to *enhance* the overall disclosure in your filing.”

### 7.1.2. Univariate validation tests for qualitative disclosure attributes

Prior to testing our formal hypotheses, in order to provide further evidence supporting the construction of *QDF* beyond our review of select comment letters (see Appendix B), we examine the univariate change in each of the five qualitative disclosure attributes between the subject and subsequent annual report. Panel A of Table 5 presents the univariate changes for the comment letter sample unrestricted by control variable availability. The univariate results provide a baseline for the tests in Panel B and validation of the

**Table 6**

The attenuating effect of rule 406 confidential treatment requests.

$$QDF = \beta_0 + \beta_1 CL + \beta_2 Post + \beta_3 CTR406 + \beta_4 CL*Post + \beta_5 CL*CTR406 + \beta_6 Post*CTR406 + \beta_7 CL*Post*CTR406 + \beta_x Controls + \varepsilon$$

| Variables      | QDF                  | QDF                   |
|----------------|----------------------|-----------------------|
| Intercept      | −3.465***<br>(−5.86) | −9.739***<br>(−16.20) |
| CL             |                      | −0.057***<br>(−2.70)  |
| Post           | 0.059***<br>(3.63)   | −0.029*<br>(−1.94)    |
| CTR406         | 0.191<br>(1.31)      | 0.062***<br>(2.78)    |
| CL*Post        |                      | −0.350***<br>(−2.68)  |
| CL*CTR406      |                      | 0.485**<br>(2.03)     |
| Post*CTR406    | −0.042<br>(−0.45)    | 0.196*<br>(1.72)      |
| CL*Post*CTR406 |                      | −0.274*<br>(−1.70)    |
| N              | 5804                 | 11,770                |
| Controls       | Yes                  | Yes                   |
| R-squared      | 0.282                | 0.303                 |

Table 6 presents regression results on the effect of a Rule 406 confidential treatment request (*CTR406*) on the association between the SEC comment letter process and the change in the qualitative disclosure factor (*QDF*). Column 1 presents the results for the comment letter sample. Column 2 presents the results of the matched, pre-post model. t-statistics are shown in parentheses below the regression coefficients and are computed using standard errors clustered by firm. All regressions include industry fixed effects. Please refer to Appendix C for variable definitions.

\* Denotes statistical significance at the 0.10 significance level.

\*\* Denotes statistical significance at the 0.05 significance level.

\*\*\* Denotes statistical significance at the 0.01 significance level.

direction and significance of the qualitative disclosure improvements as evidenced by the decrease in *Tone* and the increase in *FLSI*. However, the significant univariate increase in *Readability* is contrary to our prediction, implying the comment letter reviews appear to decrease overall annual report readability.<sup>34</sup>

In Panel B, we condition the sample based on the nature of the comment letter request as appropriate for each qualitative disclosure attribute. For *Length* we evaluate the subsample of comment letters that explicitly request a revision to a future filing and find a significant increase. For *Readability* we condition the sample on comment letters containing the keywords “plain english”, “boilerplate”, “clarify”, “explain”, or “define” and find that the annual report becomes less readable. For *Tone* we condition the sample on comment letters containing the keywords “tone”, “balanced”, “optimistic”, “positive”, or “negative” and find a significant decline in the tone of the annual report. For *NI* we condition the sample on comment letters containing the keywords “quantify” or “quantitative” and find a significant increase in the numeric intensity of the annual report. Finally, for *FLSI* we condition the sample on comment letters containing the keywords “forward-looking”, “forward looking”, “prospective”, or “future” and find a significant increase in the forward-looking intensity of the annual report. To summarize, with the exception of readability, Panel B of Table 5 provides evidence supporting the validity of our construction of *QDF* in line with the stated objectives of the SEC and the specific requests found in comment letters.

## 7.2. Confidential treatment requests (H1) and registrant negotiation (H2)

Table 6 presents regression results from estimating Eq. (2). The coefficient of interest,  $\beta_7$ , on the interaction term  $CL*Post*CTR406$  in Eq. (2) reflects the differential improvement to the qualitative disclosure factor (*QDF*) for reviews where the firm requests confidential treatment for required material disclosures under Rule 406. We find a negative and statistically significant coefficient on  $CL*Post*CTR406$  in Column 2. The result suggests that, while firms modify their qualitative disclosure consequent to a comment letter review on average, this practice is attenuated for firms making Rule 406 confidential treatment requests. The result suggests that the increase in *QDF* is mitigated by Rule 406 confidential treatment requests during the comment letter process, which provides evidence in support of H1. In untabulated analysis, we do not find a significant attenuating effect of Rule 83 CTRs, which do not relate to required disclosures, but only to supplemental information requested by the SEC staff to facilitate the filing review process.

Table 7 presents regression results from estimating Eq. (3). The coefficient of interest,  $\beta_7$ , on the interaction terms

<sup>34</sup> We note that when we omit readability from the composite disclosure measure, the inferences from our results are unchanged. In addition, in untabulated analysis, we include business and geographic segments in order to control for operational complexity that may give rise to more complex disclosures. We continue to find a statistically significant decrease in readability, which could suggest a tradeoff whereby a requested disclosure improvement along one qualitative dimension may come at the cost of lower readability.

**Table 7**  
The attenuating effect of registrant negotiation.

| $QDF = \beta_0 + \beta_1 CL + \beta_2 Post + \beta_3 Negotiation + \beta_4 CL*Post + \beta_5 CL*Negotiation + \beta_6 Post*Negotiation + \beta_7 CL*Post*Negotiation + \beta_x Controls + \varepsilon$ |                                  |                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------------------------|
| Variables                                                                                                                                                                                              | QDF                              | QDF                               |
| Intercept                                                                                                                                                                                              | −3.408 <sup>***</sup><br>(−5.44) | −9.739 <sup>***</sup><br>(−16.07) |
| CL                                                                                                                                                                                                     |                                  | −0.074 <sup>***</sup><br>(−3.12)  |
| Post                                                                                                                                                                                                   | 0.086 <sup>***</sup><br>(3.84)   | −0.032 <sup>*</sup><br>(−1.73)    |
| Negotiation                                                                                                                                                                                            | 0.118 <sup>***</sup><br>(2.98)   | 0.004<br>(0.11)                   |
| CL*Post                                                                                                                                                                                                |                                  | 0.086 <sup>***</sup><br>(3.16)    |
| CL*Negotiation                                                                                                                                                                                         |                                  | 0.068<br>(1.55)                   |
| Post*Negotiation                                                                                                                                                                                       | −0.091 <sup>**</sup><br>(−2.23)  | 0.015<br>(0.45)                   |
| CL*Post*Negotiation                                                                                                                                                                                    |                                  | −0.085 <sup>+</sup><br>(−1.75)    |
| N                                                                                                                                                                                                      | 4881                             | 11,770                            |
| Controls                                                                                                                                                                                               | Yes                              | Yes                               |
| R-squared                                                                                                                                                                                              | 0.285                            | 0.303                             |

Table 7 presents regression results on the effect of registrant negotiation on the association between the SEC comment letter process and the change in the qualitative disclosure factor (*QDF*). Column 1 presents the results for the comment letter sample. Column 2 presents the results of the matched, pre-post model. t-statistics are shown in parentheses below the regression coefficients and are computed using standard errors clustered by firm. All regressions include industry fixed effects. Please refer to Appendix C for variable definitions.

\* Denotes statistical significance at the 0.10 significance level.

\*\* Denotes statistical significance at the 0.05 significance level.

\*\*\* Denotes statistical significance at the 0.01 significance level.

*CL\*Post\*Negotiation* in Eq. (3) reflects the differential improvement to the qualitative disclosure factor (*QDF*) for reviews where the firm negotiates with the SEC. Similar to the results in Column 1, we find a statistically significant change in *CL\*Post\*Negotiation* in Column 2. The result suggests that the increase in *QDF* is mitigated by the extent of registrant negotiation during the comment letter process, which provides evidence in support of H2. As firms increase the number of rounds of negotiations with the SEC, the observed changes in qualitative disclosure are diminished, which implies that firms are able to partially circumvent requests to modify their disclosure through negotiation. In sum, the evidence provided for H1 and H2 is consistent with there being circumstances under which firms choose not to comply with the SEC, perhaps for reasons associated with proprietary costs (Robinson et al., 2011).

To better understand the types of firms that make confidential treatment requests and/or negotiate with the SEC, we examine

**Table 8**  
Cross-sectional analysis: proprietary costs, confidential treatment requests, and negotiation.

| $CTR \text{ or } Negotiation = \beta_0 + \beta_1 High\ Tech + \beta_2 Competition + \beta_3 R\ \&\ D + \beta_x Controls + \varepsilon$ |                              |                              |
|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------|------------------------------|
| Variables                                                                                                                              | CTR                          | Negotiation                  |
| Intercept                                                                                                                              | 2.579<br>(0.94)              | −3.675<br>(−1.61)            |
| High Tech                                                                                                                              | 0.471 <sup>+</sup><br>(1.65) | 0.541 <sup>+</sup><br>(1.93) |
| Competition                                                                                                                            | −0.094<br>(−0.067)           | −0.135<br>(−1.05)            |
| R & D                                                                                                                                  | 0.380 <sup>+</sup><br>(1.90) | 0.132<br>(0.69)              |
| N                                                                                                                                      | 3186                         | 2731                         |
| Controls                                                                                                                               | Yes                          | Yes                          |
| R-squared                                                                                                                              | 0.132                        | 0.125                        |

Table 8 presents cross-sectional regression results on the association between proprietary costs, confidential treatment requests, and negotiation. Column 1 presents the results for confidential treatment requests and Column 2 presents the results for negotiation. t-statistics are shown in parentheses below the regression coefficients and are computed using standard errors clustered by firm. All regressions include industry fixed effects. Please refer to Appendix C for variable definitions.

\*\* Denotes statistical significance at the 0.05 significance level.

\*\*\* Denote statistical significance at the 0.01 significance level.

\* Denotes statistical significance at the 0.10 significance level.

**Table 9**  
Changes in the QDF and firms' information environments.

| $Information\_Environment = \beta_0 + \beta_1 CL + \beta_2 Post + \beta_3 QDF + \beta_4 CL*Post + \beta_5 CL*QDF + \beta_6 Post*QDF + \beta_7 CL*Post*QDF + \beta_x Controls + \varepsilon$ |                                |                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------------------|
| Variables                                                                                                                                                                                   | BAS                            | BAS                              |
| Intercept                                                                                                                                                                                   | 0.054 <sup>***</sup><br>(3.29) | 0.040 <sup>**</sup><br>(2.09)    |
| CL                                                                                                                                                                                          |                                | −0.001<br>(−1.01)                |
| Post                                                                                                                                                                                        | 0.002 <sup>***</sup><br>(3.11) | 0.001 <sup>**</sup><br>(2.10)    |
| QDF                                                                                                                                                                                         | 0.002 <sup>**</sup><br>(2.37)  | 0.002 <sup>**</sup><br>(2.08)    |
| CL*Post                                                                                                                                                                                     |                                | −0.000<br>(−0.09)                |
| CL*QDF                                                                                                                                                                                      |                                | 0.001<br>(0.71)                  |
| Post*QDF                                                                                                                                                                                    | −0.001 <sup>*</sup><br>(−1.90) | 0.001<br>(1.02)                  |
| CL*Post*QDF                                                                                                                                                                                 |                                | −0.002 <sup>***</sup><br>(−2.67) |
| N                                                                                                                                                                                           | 5773                           | 11,762                           |
| Controls                                                                                                                                                                                    | Yes                            | Yes                              |
| R-squared                                                                                                                                                                                   | 0.045                          | 0.051                            |

Table 9 presents regression results on the association between the change in the qualitative disclosure factor (*QDF*) and the firm's information environment. Column 1 presents the results for the comment letter sample. Column 2 presents the results of the matched, pre-post model. t-statistics are shown in parentheses below the regression coefficients and are computed using standard errors clustered by firm. All regressions include industry fixed effects. Please refer to Appendix C for variable definitions.

- \* Denotes statistical significance at the 0.10 significance level.
- \*\* Denotes statistical significance at the 0.05 significance level.
- \*\*\* Denotes statistical significance at the 0.01 significance level.

**Table 10**  
Changes in the QDF and information intermediary following.

| $Information\_Intermediary\_Following = \beta_0 + \beta_1 CL + \beta_2 Post + \beta_3 QDF + \beta_4 CL*Post + \beta_5 CL*QDF + \beta_6 Post*QDF + \beta_7 CL*Post*QDF + \beta_x Controls + \varepsilon$ |                                   |                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-----------------------------------|
| Variables                                                                                                                                                                                               | Analyst_Cov                       | Analyst_Cov                       |
| Intercept                                                                                                                                                                                               | −10.913 <sup>***</sup><br>(−6.02) | −15.742 <sup>***</sup><br>(−6.65) |
| CL                                                                                                                                                                                                      |                                   | −0.079<br>(−1.01)                 |
| Post                                                                                                                                                                                                    | 0.186 <sup>***</sup><br>(4.49)    | 0.193 <sup>***</sup><br>(4.36)    |
| QDF                                                                                                                                                                                                     | −0.110<br>(−1.34)                 | 0.151 <sup>**</sup><br>(2.02)     |
| CL*Post                                                                                                                                                                                                 |                                   | 0.060<br>(1.02)                   |
| CL*QDF                                                                                                                                                                                                  |                                   | −0.183 <sup>**</sup><br>(−2.02)   |
| Post*QDF                                                                                                                                                                                                | 0.124 <sup>**</sup><br>(2.09)     | 0.160 <sup>**</sup><br>(2.36)     |
| CL*Post*QDF                                                                                                                                                                                             |                                   | −0.112<br>(−1.31)                 |
| N                                                                                                                                                                                                       | 5110                              | 10,432                            |
| Controls                                                                                                                                                                                                | Yes                               | Yes                               |
| R-squared                                                                                                                                                                                               | 0.539                             | 0.534                             |

Table 10 presents regression results on the association between the change in the qualitative disclosure factor (*QDF*) and following by information intermediaries. Column 1 presents the results for the comment letter sample. Column 2 presents the results of the matched, pre-post model. t-statistics are shown in parentheses below the regression coefficients and are computed using standard errors clustered by firm. All regressions include industry fixed effects. Please refer to Appendix C for variable definitions.

- \* Denotes statistical significance at the 0.10 significance level.
- \*\* Denotes statistical significance at the 0.05 significance level.
- \*\*\* Denotes statistical significance at the 0.01 significance level.

**Table 11**  
Changes in the QDF and litigation risk.

| $Litigation = \beta_0 + \beta_1 CL + \beta_2 \Delta QDF + \beta_3 CL * \Delta QDF + \beta_x Controls + \varepsilon$ |                         |                         |
|---------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------------|
| Variables                                                                                                           | Litigation              | Litigation              |
| Intercept                                                                                                           | – 14.322***<br>(– 4.07) | – 13.495***<br>(– 3.81) |
| CL                                                                                                                  | – 0.490***<br>(– 3.53)  | – 0.303*<br>(– 1.83)    |
| $\Delta QDF$                                                                                                        |                         | 0.547***<br>(2.95)      |
| $CL * \Delta QDF$                                                                                                   |                         | – 0.633**<br>(– 2.12)   |
| N                                                                                                                   | 5567                    | 5567                    |
| Controls                                                                                                            | Yes                     | Yes                     |

Table 11 presents logistic regression results on the probability of a class-action lawsuit (*Litigation*) following a comment letter review and its association with changes in the qualitative disclosure factor ( $\Delta QDF$ ). t-statistics are shown in parentheses below the regression coefficients and are computed using standard errors clustered by firm. All regressions include industry fixed effects. Please refer to Appendix C for variable definitions.

\* Denotes statistical significance at the 0.10 significance level.

\*\* Denotes statistical significance at the 0.05 significance level.

\*\*\* Denotes statistical significance at the 0.01 significance level.

whether proprietary costs are a driver of firms' reluctance to be forthcoming in their disclosures (Verrecchia, 1983). We identify firms likely to face proprietary cost concerns in several ways. Following prior literature (Bushee et al. (2003), Li et al. (2013), and Wang (2007)) we examine firms in high tech industries, firm competition, and R & D intensive firms, respectively. In all cases (High Tech, Competition, and R & D), such firms are likely to possess trade secrets about their business strategies and plans that would be costly if publicly disclosed. Consistent with such concerns, in Table 8 we present cross-sectional analysis and find in Column 1 that firms in high tech industries and with greater amounts of R & D are more likely to request confidential treatment. In addition, in Column 2 we find that firms in high tech industries are also more likely to negotiate.

### 7.3. Disclosure improvements and firms' information and litigation environments (H3 - H5)

Tables 9 and 10 present regression results from estimating Eq. (4) regarding firms' information environments. The coefficient of interest,  $\beta_7$ , on the interaction terms  $CL * Post * QDF$  in Eq. (4) reflects the change in the firm's bid-ask spread or information intermediary following, respectively, for firms that improve their disclosure following a comment letter review. In Table 9, consistent with H3, the negative coefficient on  $CL * Post * QDF$  in Column 2 indicates that the disclosure improvements appear to decrease information asymmetry immediately following the filing of the subsequent annual report. Similar inferences are obtained from Column 1.<sup>35</sup> In Table 10, consistent with H4, the positive coefficient on  $Post * QDF$  in Column 1 suggests that disclosure improvements increase analyst following. However, the result from the matched, pre-post design found in Column 2 (coefficient on  $CL * Post * QDF$ ) is not statistically significant at conventional levels.

Lastly, Table 11 presents regression results from estimating Eqs. (5) and (6) regarding firms' litigation risk. In Column 1 of Table 11, the negative coefficient on  $CL$  suggests that firms undergoing a review experience lower litigation risk following a review. Further, consistent with H5, Column 2 of Table 11 indicates that firms with the largest changes in  $QDF$  experience a greater reduction in litigation risk. Collectively, the results suggest that SEC comment letter reviews facilitate improvements in firms' disclosures which impact firms' information and legal environments. As such, we interpret these results as being indicative of firms' commitment to more transparent disclosure following a comment letter review which translates into better information environments for firm stakeholders as well as lower litigation risk for the firm.

## 8. Conclusion

Prior academic research has called for more empirical research on the effectiveness of disclosure regulation; however, little research has been done to assess the impact of the SEC comment letter process on firm disclosures and the implications of that process for firms' information environments. Given the SEC's objective of enhancing informational transparency for investors, an examination of the nature, extent, and impact of improvements to firms' disclosures coinciding with the comment letter process is of interest to academics and regulators, as well as to practitioners and standard setters. Considering the general interpretive guidance that the SEC provides to firms, comment letter reviews may carry little weight and therefore lead to inconsequential disclosure changes, or even to no changes at all, suggesting that SEC resources can be better used for other purposes. In addition, firms can choose to avoid making

<sup>35</sup> Johnston and Petacchi (2017) also finds a decrease in information asymmetry based on the evidence of a decrease in the quarterly average of a measure of bid-ask spread in the eight quarters following a comment letter review.

substantive disclosure changes by making a confidential treatment request, negotiating with the SEC staff, or failing to fully comply with the staff's requests.

The initial evidence presented suggests that comment letter reviews are associated with improvements in qualitative disclosure. Our univariate tests generally provide corroborating evidence that supports our multivariate analysts in that comment letters are associated with revised disclosures which are (i) expanded, (ii) less optimistic, (iii) more quantitative, and (iv) more prospective. We then provide evidence of an attenuating effect on disclosure improvements following a comment letter review for firms that submit a Rule 406 confidential treatment request. Consistent with proprietary cost concerns, we find that firms in high tech industries and with greater amounts of R & D are more likely to request confidential treatment. Next, we provide evidence which shows that firms' propensity to negotiate with the SEC regarding comment letter requests also has an attenuating effect on disclosure improvements. Inconsistent with the SEC's stated intentions of the filing review process, these results suggest that the comment letters do not enhance disclosure under certain circumstances.

We then turn to examine informational transparency in order to more directly test the SEC's stated intention behind the filing review process. Our results reveal that qualitative disclosure changes prompted by SEC comment letters decrease information asymmetry and, to some extent, increase analyst following. In addition, our results suggest that firms' commitment to facilitate more transparent disclosure following a comment letter review reduces firms' litigation risk. Hence, rather than creating information or legal uncertainty, these results suggest that firms that commit to provide more transparent disclosure, i.e., those not "pushing back" on the SEC, benefit from the comment letter process.

Our study builds on prior work by providing evidence on the role the SEC comment letter process has on enhancing corporate disclosure and its implications for firms' information and legal environments. Consistent with the calls by Healy and Palepu (2001), Leuz and Wysocki (2016), and Beyer et al. (2010), we add to the literature on disclosure regulation by documenting the effectiveness of the interplay between regulators and registrants in the disclosure regulation compliance process. Collectively, our results are consistent with the SEC's objective of influencing firm disclosures as its "chief regulatory tool" in order to enhance informational transparency. As such, the study speaks to the "...incentives to produce and disseminate information and the resulting information asymmetry between insiders and outsiders and among investors..." (Beyer et al., 2010).

## Appendix A. Examples of comment and response letter types

### A.1. Non-response example

#### *American Oil & Gas, Inc. comment letter dated March 23, 2016*

We issued a comment on the above captioned filing on January 14, 2016. On March 8, 2016, we issued a follow-up letter informing you that this comment remained outstanding and unresolved, and absent a substantive response, we would act consistent with our obligations under the federal securities laws. As you have not provided a substantive response, we are terminating our review and will take further steps as we deem appropriate. These steps include releasing publicly, through the agency's EDGAR system, all correspondence, including this letter, relating to the review of your filing, consistent with the staff's decision to publicly release comment and response letters relating to disclosure filings it has reviewed.

### A.2. Non-compliance example

#### *Stanley Black & Decker, Inc. comment letter dated March 30, 2012*

We note that in your letter dated April 8, 2010 in response to comment 1 of our letter dated March 30, 2010, you stated that you will disclose company sponsored research and development expenditures in Item 1 of future filings in accordance with Item 101(c)(1) (xi) of Regulation S-K. However, it does not appear that you have done so. Please comply with this comment in future filings.

### A.3. Negotiation examples

#### *Myriad Genetics, Inc. response letter dated March 12, 2008*

The specific and detailed disclosure of research and development costs incurred by us, as requested, especially in light of the lack of equivalent disclosure by many of our competitors, would place us at an unfair competitive disadvantage .... Accordingly, we respectfully propose that we be allowed to continue reporting research and development costs on an aggregate basis.

#### *Cedar Fair, L.P. response letter dated March 29, 2013*

The Partnership is filing a Request for Reconsideration with the Office of the Chief Accountant in reference to this Comment. (This appeal was repeated three times in this response letter – half of the six outstanding comments).

### A.4. Confidential treatment request ("CTR") Examples

#### *PepsiCo, Inc. response letter dated October 8, 2013*

##### Rule 83 Confidential Treatment Request by PepsiCo, Inc. Request #1

In the event we decide to pursue new business opportunities in any of these countries our activities would be conducted in accordance with U.S. laws, including where necessary licenses issued by the appropriate regulatory authority. [\*\*].

PepsiCo, Inc. respectfully requests that the information contained in Request #1 be treated as confidential information and that

the Commission provide timely notice ... before it permits any disclosure of the bracketed information contained in Request #1.

*The Valspar Corporation response letter dated February 25, 2016*

Even if the Staff disagrees with our determination regarding the materiality of the EPS Targets, we nevertheless respectfully submit that the EPS Targets constitute confidential information of the Company, the disclosure of which would result in competitive harm to our business. C & DI 118.04 and Instruction 4 to Item 402(b) of Regulation S-K provide that “registrants are not required to disclose target levels with respect to specific quantitative or qualitative performance related factors considered by the compensation committee or the board of directors, or any other factors or criteria involving confidential trade secrets or confidential commercial or financial information, the disclosure of which would result in competitive harm for the registrant.” Instruction 4 goes on to state that “[t]he standard to use when determining whether disclosure would cause competitive harm for the registrant is the same standard that would apply when a registrant requests confidential treatment of confidential trade secrets or confidential commercial or financial information pursuant to Securities Act Rule 406 and Exchange Act Rule 24 b-2, each of which incorporates the criteria for non-disclosure when relying upon Exemption 4 of the Freedom of Information Act...

## Appendix B. Examples of comment letters requests

### B.1. Length example

*Numerex Corporation comment letter dated August 15, 2013*

In future filings, please expand your discussion of your results of operations to address why your operating expenses are increasing at a higher rate and the steps management is taking, or plans to take, to counter this apparent negative trend.

### B.2. Readability example

*Scholastic Corporation comment letter dated February 7, 2012*

We believe your disclosure would be much more useful to investors if it was focused on describing the current status of the matter, in plain English, rather than on the history of legal proceedings. Provide us with a copy of your intended revised disclosure. Refer to the disclosure requirements set forth by paragraphs 1–5 of ASC 450–20–50.

### B.3. Tone example

*iParty Corporation comment letter dated March 9, 2005*

In addition, the tone of the Overview section is overwhelming positive despite the significant negative evidence you cite in your response to comment 7. In future filings in addition to providing increased disclosure of the change in income taxes and the related valuation allowance in your discussion of results of operations please provide a balanced discussion in the Overview section of the positive and negative factors that have impacted your company for the historical periods presented.

### B.4. Numerical example

*Horizon Pharma, Inc. comment letter dated August 13, 2013*

We believe that your disclosure herein related to estimates of items that reduce gross revenue such as product returns, chargebacks, customer rebates and other discounts and allowances could be improved. Please provide us proposed disclosure to be included in future periodic reports to address the following...To the extent that information you consider in the preceding bullet is quantifiable, disclose both quantitative and qualitative information and to what extent information is from external sources.

### B.5. Forward-looking statement example

*Moody's Corporation comment letter dated July 26, 2012*

In future filings, please provide more specific detail as to why line items have changed and whether any changes have prospective implications. For instance, if you anticipate changes in your fee types and the addition of new fee initiatives and price increases in the future, please specifically discuss the products or services within a specific operating segment these fee increases and changes are associated with. Please also note in addressing prospective financial condition and operating performance, there are circumstances, particularly regarding known material trends and uncertainties, where forward-looking information is required to be disclosed.

## Appendix C. Variable definitions

**Age** – firm age as of the most recent fiscal quarter prior to the comment letter calculated as years listed on Compustat.

**Analyst\_Cov** – average number of analysts covering the firm in the 30 days *following* the annual report filing date based on the number of outstanding forecasts.

**Analysts** – number of analysts following the firm as of the most recent fiscal quarter *prior* to the comment letter based on the number of outstanding forecasts.

**BAS** – abnormal bid-ask-spread calculated as the event period average daily percent spread minus the pre-period average daily

percent spread. The event period is trading days (0, +2) relative to the annual report filing date and the pre-period is the 45 trading days prior. Daily percent spread is calculated as  $((\text{high offer price} - \text{low bid price}) / \text{midpoint}) * 100$ .

**Beta** – market beta calculated from regressions of daily firm returns on the market index for the fiscal year prior to the comment letter.

**BTM** – book to market value of equity as of the most recent fiscal quarter prior to the comment letter.

**Competition** – indicator variable for companies with above-median competition following Li et al. (2013).

**CTR** – indicator variable for a confidential treatment request by the registrant as part of the comment letter process.

**CTR406** – indicator variable for a confidential treatment request that explicitly invokes Rule 406 of the Securities Act of 1933, which is related to disclosures required to be filed with the SEC.

**CL** – indicator variable for the annual report that is the subject of a comment letter review.

**CL\_Complexity** – a measure of the quantity and significance of accounting comments raised during the review expressed as the total number of words in the initial SEC comment letter.

**Disclosure\_Trend** – log of industry-average annual report word count using the 4-digit SIC code.

**FLSI** – a measure of the amount of forward-looking disclosure, expressed as the number of forward-looking sentences contained within the annual report following Li (2010).

**High Tech** – indicator variable for companies in high tech industries as classified by Bushee et al. (2003).

**IO** – institutional ownership as a percentage as of the most recent quarter prior to the comment letter.

**Length** – a measure of the amount of disclosure, expressed as the number of sentences contained within the annual report.

**Litigation** – indicator variable for a firm undergoing a securities class-action lawsuit within a year following the completion of a comment letter review.

**LogAssets** – logged total assets as of the most recent fiscal quarter prior to the comment letter.

**LogSize** – logged market value of equity as of the most recent fiscal quarter prior to the comment letter.

**Loss** – indicator variable for negative earnings in the most recent fiscal quarter prior to the comment letter.

**M & A** – indicator variable for a merger or acquisition involving the firm as either an acquirer or a target in the year of or year prior to the annual report being reviewed.

**Negotiation** – indicator variable for above-median number of comment letter rounds. Rounds is calculated as the number of letters issued by the SEC during a review, excluding the “Completion of Review” letter.

**NI** – a measure of numerical intensity, expressed as the count of non-date numbers contained within the text of the annual report.

**Post** – indicator variable for the annual report subsequent to the report that was the subject of the comment letter review.

**QDF** – factor analysis composite of our five disclosure attributes (*Length*, *Readability*, *Tone*, *NI*, and *FLSI*). Note we multiply *Readability* and *Tone* by -1 before computing the factor in order to consistently interpret an increase as an improvement in the disclosure.

**ΔQDF** – indicator variable for the highest quartile change in *QDF*.

**R & D** – indicator for companies with high R & D intensity, measured as above-median R & D expenditures scaled by total assets.

**Readability** – the Fog Index of the annual report expressed in the number of years of education required to understand a passage of text and computed as  $0.4[(\text{words/sentences}) + 100(\text{three or more syllable words/words})]$ . Note that a higher Fog Index indicates the disclosure is less readable.

**Restate** – indicator variable for a restatement in the year of or year prior to the annual report being reviewed.

**ROA** – return on assets as of the most recent fiscal quarter prior to the comment letter.

**Stdev\_Earn** – earnings volatility calculated as the standard deviation of the 12 quarterly earnings prior to the comment letter.

**Tone** – a measure of optimism expressed as the number of optimistic words less pessimistic words contained within the annual report using the dictionaries developed by Henry (2008) and Henry and Leone (2015) for use in financial discourse.

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